NED University of Engineering and Technology, Karachi

Computer Science and Information Technology

COMPLEX COMPUTING PROBLEM (CCP)
PROJECT SUBMISSION



Knowledge-Based Systems (CT-359)

Project Title	Intelligent Knowledge-Based System for Industrial Equipment Fault Diagnosis and Maintenance Decision Support (FaultSense AI)
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Submitted To:

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PROBLEM STATEMENT:

Industrial equipment fault diagnosis is a complex process because machine failures can result from varying operating conditions, abnormal sensor readings, and underlying mechanical or electrical issues. Identifying these faults requires more than detecting abnormal readings; it also involves understanding the relationships between equipment, machine components, symptoms, fault categories, their possible causes, and appropriate maintenance actions. A Knowledge-Based System can address this challenge by representing such industrial knowledge in a structured form, allowing diagnostic information and maintenance procedures to be organized and used systematically.

FaultSense AI addresses this problem through a structured, dictionary-based symbolic knowledge base that stores fault-related information, including affected components, diagnostic conditions, technical definitions, ISO references, and role-based maintenance actions. It also considers the requirements of different stakeholders: floor operators need immediate fault alerts and safety guidance, maintenance technicians require detailed fault explanations and troubleshooting instructions, while plant managers need information about downtime, financial impact, and maintenance priorities. Practical constraints such as sensor data complexity, class imbalance, incomplete diagnostic information, safety requirements, and production continuity are also considered.

To support accurate and explainable diagnosis, FaultSense AI combines a tuned XGBoost classifier with a symbolic rule-based inference engine. The classifier identifies potential abnormal operating conditions, while the inference engine evaluates diagnostic rules and connects identified faults with relevant knowledge from the knowledge base. This hybrid approach is selected to balance predictive performance, explainability, safety, maintenance costs, and practical system requirements.

PROBLEM DESCRIPTION:

In high-precision manufacturing plants, machinery such as CNC milling machines operates under continuous thermal, mechanical, and electrical stress. Over time, machine components may experience excessive wear, coolant-related problems, electrical overloads, or thermal abnormalities, potentially leading to equipment failure and unexpected production downtime. Traditional monitoring approaches often rely on reactive maintenance or manual threshold checking, making it difficult for operators to continuously interpret sensor readings such as air temperature, process temperature, rotational speed, torque, and tool wear. Industrial fault diagnosis also involves different sources of knowledge, including machine telemetry, equipment specifications, technical fault definitions, industrial standards, and maintenance procedures. Multiple symptoms may be associated with different possible failure conditions, while incomplete or uncertain information can make accurate diagnosis more challenging. Furthermore, stakeholders have different priorities: floor operators focus on immediate safety, maintenance technicians require accurate fault information and repair guidance, and plant managers must consider maintenance costs, downtime, and production continuity.

To address these challenges, FaultSense AI provides a web-based industrial fault diagnosis system that accepts machine sensor readings and analyses them to identify abnormal operating conditions and potential fault categories. The tuned XGBoost model provides data-driven predictions, while the rule-based inference engine evaluates diagnostic conditions and connects identified faults with relevant information stored in the symbolic knowledge base. Based on the diagnostic results, the system generates reports containing fault status, confidence scores, detected fault categories, technical explanations, operational impact, and maintenance recommendations tailored to different stakeholders. This approach aims to reduce reliance on manual interpretation and provide structured diagnostic information to support timely maintenance decisions.

PROBLEM ANALYSIS:

1. PROBLEM UNDERSTANDING:

Industrial fault diagnosis requires analysing machine operating conditions to identify abnormal behaviour and possible equipment failures. FaultSense AI focuses on CNC machine conditions using five sensor parameters: air temperature, process temperature, rotational speed, torque, and tool wear. These parameters may interact in different ways, making manual interpretation and fault identification challenging.

The system processes the entered sensor readings through input validation and feature engineering before applying a tuned XGBoost classifier. Its rule-based inference engine then evaluates diagnostic conditions to identify specific fault categories. The identified fault codes are connected with the structured knowledge base to retrieve relevant component information, technical definitions, safety considerations, and maintenance actions. The main objective is to provide understandable diagnostic results rather than only predicting whether a machine is operating normally or experiencing a failure.

2. STAKEHOLDERS AND THEIR REQUIREMENTS:

FaultSense AI considers three primary stakeholders involved in industrial machine operation and maintenance:

- **Floor Operators:** Require clear machine status, immediate fault alerts, and simple safety instructions. The knowledge base provides fault-specific actions, such as emergency stopping for certain conditions, safely parking the CNC axis, or avoiding contact with electrical equipment.
- **Maintenance Technicians:** Require detailed fault information, affected component identification, technical definitions, diagnostic conditions, and troubleshooting instructions. The system provides maintenance actions such as replacing worn cutting inserts, inspecting coolant systems, checking electrical connections, and examining mechanical components.
- **Plant Managers:** Require information about safety criticality, expected downtime, financial cost, and production continuity. The knowledge base provides operational impact details and management-level actions to support maintenance scheduling, resource allocation, and production decisions.

3. CONSTRAINTS AND CONFLICTING REQUIREMENTS:

- **Safety vs. Production Continuity:** Stopping a machine can help prevent further damage and safety hazards, but it may also interrupt production. The system provides fault-specific safety instructions and operational impact information to support appropriate maintenance decisions.
- **Data Quality and Incomplete Inputs:** Sensor readings may contain missing values or fall outside expected physical ranges. The inference engine includes input validation for the five sensor parameters and checks machine type information. However, missing and invalid readings are tracked, so incomplete inputs may affect the reliability of diagnostic results.
- **Class Imbalance:** The AI4I 2020 dataset contains significantly fewer failure records than normal operating records, making it challenging for the classifier to identify minority-class failures. To address this issue, SMOTE-based oversampling and classification threshold adjustments were experimented with. However, SMOTE did not improve the overall accuracy, so model performance had to be evaluated by considering the trade-off between detecting failures and avoiding false alarms rather than relying on accuracy alone.

- **Probabilistic Predictions vs. Deterministic Rules:** The XGBoost classifier produces predictions with confidence probabilities, whereas the symbolic inference engine evaluates specific diagnostic conditions using defined thresholds. Combining both approaches helps provide data-driven predictions alongside rule-based fault identification.
- **Machine-Type Variations:** Fault thresholds may differ according to machine quality type. The knowledge base defines Lite (L), Medium (M), and High (H) profiles, and the inference engine uses type-specific limits for tool wear and overstrain detection.

4. ROOT CAUSE / REQUIREMENT IDENTIFICATION:

FaultSense AI considers five fault categories represented in its knowledge base. Their definitions and diagnostic conditions are as follows:

- **Tool Wear Failure (TWF):** Associated with abrasive friction and gradual deterioration of the cutting tool. It is detected when tool wear exceeds the machine-type-specific limit: 240 for L, 210 for M, and 220 for H.
- **Heat Dissipation Failure (HDF):** Associated with reduced cooling capacity due to possible fluid loss or blockage. It is detected when the temperature difference between process and air temperature is below 2.0 K and rotational speed is below 1380 RPM.
- **Power Failure (PWF):** Associated with abnormal mechanical power conditions that may indicate electrical drive or motor problems. It is detected when calculated mechanical power falls below 3500 W or exceeds 9000 W.
- **Overstrain Failure (OSF**): Associated with excessive mechanical stress caused by the combination of tool wear and torque. It is detected when the product of tool wear and torque exceeds the machine-type-specific limit: 11000 for L, 12000 for M, and 13000 for H.
- **Random Failure (RNF):** Used as a general fallback category when the ML model predicts a failure but none of the specific diagnostic rules is triggered. It represents an unmatched anomaly rather than a separately verified physical root cause.

Based on these diagnostic requirements, the system includes input validation, feature engineering, a tuned XGBoost classifier, an independent rule-based inference engine, a structured symbolic knowledge base, and an explanation generator. These components work together to identify potential faults and provide relevant technical information and stakeholder-specific maintenance recommendations.

5. ASSUMPTIONS:

- Machine sensor readings, including air temperature, process temperature, rotational speed, torque, and tool wear, are available through manual user input or the backend API.
- Industrial machines are represented using three predefined quality types: Lite (L), Medium (M), and High (H), as supported by the system.
- The entered sensor readings are assumed to represent the machine's operating condition at the time of diagnosis.
- The diagnostic thresholds and machine-type-specific limits stored in the knowledge base are treated as predefined conditions for identifying potential faults.
- The technical definitions, ISO references, safety information, downtime estimates, and maintenance protocols stored in the knowledge base are assumed to be relevant to the represented fault categories.
- The XGBoost model is assumed to provide useful predictions for inputs that are reasonably similar to its training data. Its predictions and confidence scores are considered decision-support information rather than guaranteed confirmation of actual machine failure.

KNOWLEDGE REPRESENTATION AND KBS DESIGN:

1. SELECTED KNOWLEDGE REPRESENTATION FORMALISM:

FaultSense AI uses a Hybrid Knowledge Representation Approach that combines symbolic knowledge representation, production rules, and machine learning.

- **Symbolic Knowledge Representation:** Industrial knowledge is stored in a structured Python dictionary (KNOWLEDGE_BASE). It contains fault definitions, affected components, ISO references, machine-type limits, maintenance actions, and operational constraints.
- **Production Rules:** IF-THEN rules evaluate machine sensor readings against predefined diagnostic conditions to identify specific fault categories.
- **Machine Learning:** A tuned XGBoost classifier analyses sensor data and predicts potential machine failures.

This hybrid approach combines structured industrial knowledge and forward-chaining rule-based reasoning with data-driven predictions to support fault diagnosis and maintenance recommendations.

2. KNOWLEDGE BASE:

FaultSense AI uses a dictionary-based symbolic knowledge base implemented in knowledge_base.py. Each fault code acts as a primary key and is associated with relevant technical and maintenance information.

The knowledge base contains:

- **Fault Identification:** Unique fault IDs and failure mode names.
- **Target Components:** Machine components associated with each fault.
- **Technical Definitions**: Fault descriptions, symptoms, and diagnostic information.
- **ISO References:** Relevant industrial standard references associated with fault categories.
- **Machine-Type Profiles:** Diagnostic limits for Lite (L), Medium (M), and High (H) machine types.
- **Action Protocols:** Maintenance instructions for floor operators, maintenance technicians, and plant managers.
- **Operational Constraints:** Safety criticality, estimated downtime, financial cost index, and production continuity impact.

Fault Modes in the Knowledge Base:

Fault Code	Fault Mode	Description
TWF	Tool Wear Failure	Excessive wear and deterioration of the cutting tool.
HDF	Heat Dissipation Failure	Abnormal cooling conditions affecting heat dissipation.
PWF	Power Failure	Abnormal mechanical power conditions.
OSF	Overstrain Failure	Excessive mechanical stress caused by the combination of tool wear and torque.
RNF	Random Failure	General fallback category for an ML-detected anomaly without a matching specific rule.

Diagnostic Rules:

Fault Code	Detection Condition
TWF	Tool wear exceeds the machine-type-specific limit.
HDF	Temperature difference (Process Temp – Air Temp) is below 2.0 K AND rotational speed is below 1380 RPM.
PWF	Mechanical power is below 3500 W OR above 9000 W.
OSF	Tool Wear × Torque exceeds the machine-type-specific limit.

The machine-type-specific limits used by the rules are:

Machine Type	TWF Limit	OSF Limit
Lite (L)	240	11000
Medium (M)	210	12000
High (H)	220	13000

3. INFERENCE MECHANISM:

FaultSense AI uses a Forward Chaining Inference Mechanism integrated with an XGBoost classifier. The inference process follows these steps:

- **Input Validation:** The system receives machine sensor readings and checks the input values for missing or invalid data.
- **Feature Engineering:** It calculates additional features such as temperature difference, mechanical power, overstrain index, and torque-speed ratio.
- **ML Classification:** The tuned XGBoost model analyses the processed sensor data and predicts whether the machine is normal or faulty, along with a confidence score.
- **Forward Chaining:** The inference engine evaluates sensor facts against predefined IF-THEN diagnostic rules to identify specific faults such as TWF, HDF, PWF, and OSF.
- **Result Fusion:** The ML prediction and rule-based results are combined to determine the final machine status. If the ML model predicts a fault but no specific rule matches, RNF is assigned.
- **Knowledge Retrieval**: The system retrieves relevant fault definitions, affected components, ISO references, and maintenance actions from the knowledge base.

In this process, known sensor facts trigger the diagnostic rules, which produce fault conclusions based on the defined conditions.

4. CONFLICT RESOLUTION STRATEGY:

FaultSense AI uses a rule-priority and result-fusion strategy to determine the final diagnostic status.

- **Rule Priority:** If any diagnostic rule identifies a fault, the overall machine status is set to Fault Detected, even if the ML classifier predicts normal operation.
- **ML-Based Fallback:** If the ML model predicts a fault but no specific diagnostic rule matches, the system assigns RNF.
- **Multiple Fault Detection:** If multiple diagnostic rules are triggered, all matching fault codes are collected in the diagnostic result.
- **Normal Status:** If neither the ML model nor the symbolic rules detect a fault, the system returns a Normal status.

This strategy allows the system to consider both machine learning predictions and rule-based diagnostic results while retaining multiple detected fault categories.

5. EXPLANATION FACILITY:

FaultSense AI includes an explanation generator that converts diagnostic results into human-readable reports. It uses the retrieved knowledge-base information to explain detected faults and provide relevant maintenance guidance.

The explanation facility presents information through five main sections:

- **What:** Identifies the detected fault code, failure mode, and affected component.
- **Why:** Explains the diagnostic condition behind the detected fault using relevant sensor readings and predefined thresholds.
- **How Sure:** Displays the prediction confidence percentage generated by the XGBoost model.
- **Impact:** Presents information about safety criticality, expected downtime, financial cost, and production impact.
- **Action Protocols:** Provides role-based maintenance recommendations for floor operators, maintenance technicians, and plant managers.

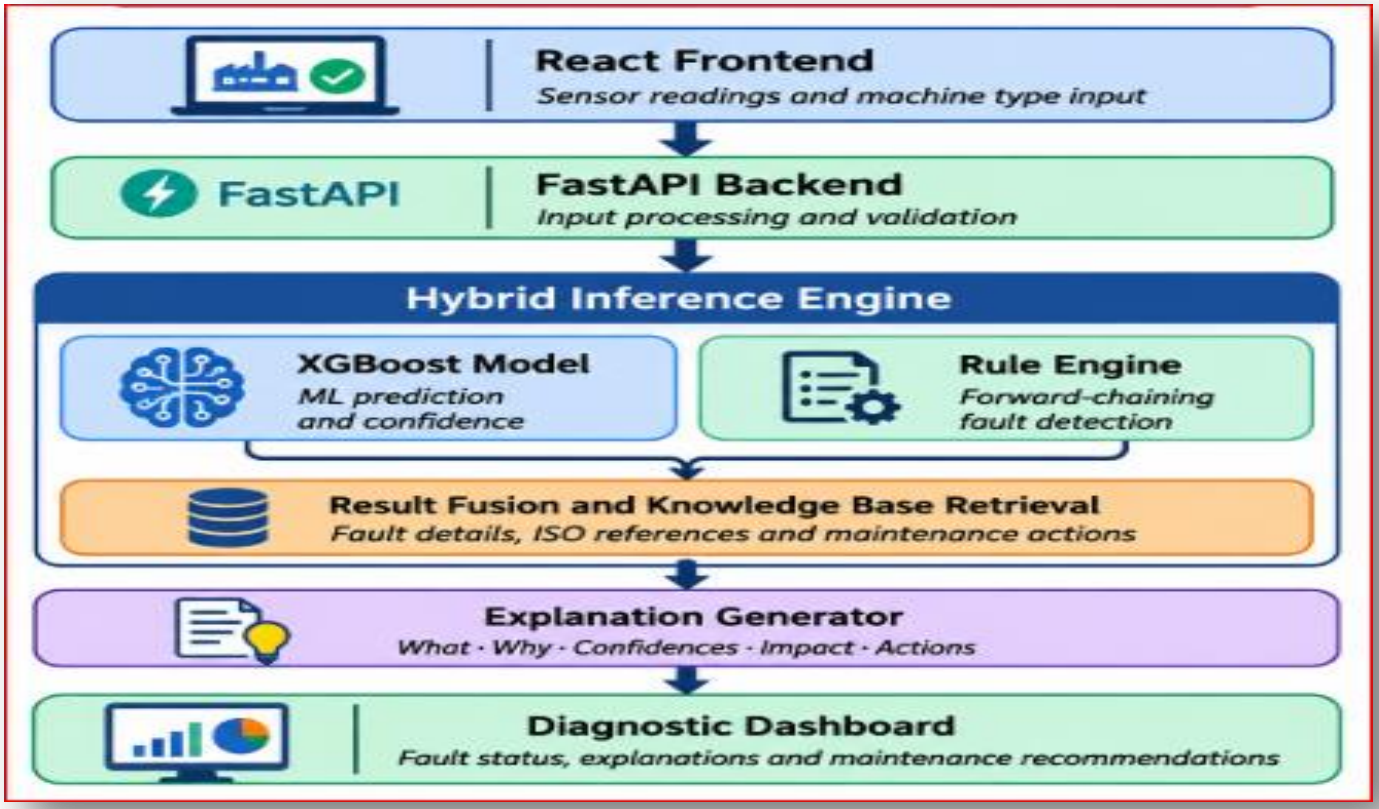
For normal machine conditions, the system displays a normal-status explanation. When a fault is detected, it generates a diagnostic report to help stakeholders understand the result and take appropriate action.

6. SYSTEM ARCHITECTURE:

FaultSense AI follows a modular architecture consisting of a frontend interface, FastAPI backend, hybrid inference engine, knowledge base, trained ML model, and explanation generator.

The system operates through the following stages:

- **Telemetry Input:** The user enters machine sensor readings and selects the machine type through the React-based frontend.
- **Backend Processing:** The frontend sends the input to the FastAPI backend through the /predict endpoint.
- **Validation and Feature Engineering:** The inference engine validates the readings and calculates the required derived features.
- **Hybrid Inference:** The tuned XGBoost model predicts the machine condition, while the rule engine evaluates diagnostic conditions to identify specific fault categories.
- **Knowledge Base Retrieval:** The detected fault codes are used to retrieve technical definitions, affected components, ISO references, operational constraints, and role-based maintenance actions.
- **Explanation and Reporting:** The explanation generator prepares a human-readable diagnostic report, which is returned to the frontend and displayed on the diagnostic dashboard.



DATASET / KNOWLEDGE SOURCES:

Source	Type	Size / Records	Purpose in FaultSense AI	Citation / URL
AI4I 2020 – Messy Dataset	Project-prepared Predictive Maintenance Dataset	10,150 rows	Data inspection, cleaning, preprocessing, feature engineering and model development.	https://www.kaggle.com/datasets/stephanmatzka/predictive-maintenance-dataset-ai4i-2020
Cai et al. (2024)	Research Paper	—	Research background for knowledge-based industrial equipment fault diagnosis.	https://link.springer.com/article/10.1007/s00170-024-12998-x
ISO 15641:2001	International Standard	—	Milling cutter safety requirements.	https://www.iso.org/standard/27337.html
ISO 13374-1:2003	International Standard	—	Machine condition monitoring and diagnostic data processing.	https://www.iso.org/standard/21832.html
IEC 60204-1:2016	International Standard	—	Electrical equipment and safety requirements of machinery.	https://webstore.iec.ch/en/publication/26037
ISO 3408-2:2021	International Standard	—	Ball screw technical specifications and dimensions.	https://www.iso.org/standard/75530.html
ISO 12100:2010	International Standard	—	Machinery safety, risk assessment and risk reduction.	https://www.iso.org/standard/51528.html

1. DATASET OVERVIEW:

FaultSense AI uses the AI4I 2020 Predictive Maintenance Dataset, available on Kaggle and the UCI Machine Learning Repository. It contains industrial machine sensor readings and failure labels used for training and evaluating the machine-learning model.

Table: Dataset Summary

Attribute	Description
Dataset Name	AI4I 2020 Predictive Maintenance Dataset
Original Records	10,000
Number of Columns	14
Data Format	CSV
Target Variable	Machine Failure (0/1)
Failure Categories	TWF, HDF, PWF, OSF, RNF
Class Distribution	9,661 Normal, 339 Failure

The dataset is highly imbalanced because only 3.39% of the original records represent machine failures. The project also contains a messy version with 10,150 rows, prepared for testing data-cleaning procedures.

AI4I 2020 Dataset Preview from Kaggle:

	A	B	C	D	E	F	G	H	I	J	K	L	M	N
1	UDI	Product ID	Type	Air temper	Process te	Rotational	Torque [Nr	Tool wear	Machine fa	TWF	HDF	PWF	OSF	RNF
2	1	M14860	M	298.1	308.6	1551	42.8	0	0	0	0	0	0	0
3	2	L47181	L	298.2	308.7	1408	46.3	3	0	0	0	0	0	0
4	3	L47182	L	298.1	308.5	1498	49.4	5	0	0	0	0	0	0
5	4	L47183	L	298.2	308.6	1433	39.5	7	0	0	0	0	0	0
6	5	L47184	L	298.2	308.7	1408	40	9	0	0	0	0	0	0
7	6	M14865	M	298.1	308.6	1425	41.9	11	0	0	0	0	0	0
8	7	L47186	L	298.1	308.6	1558	42.4	14	0	0	0	0	0	0
9	8	L47187	L	298.1	308.6	1527	40.2	16	0	0	0	0	0	0
10	9	M14868	M	298.3	308.7	1667	28.6	18	0	0	0	0	0	0
11	10	M14869	M	298.5	309	1741	28	21	0	0	0	0	0	0
12	11	H29424	H	298.4	308.9	1782	23.9	24	0	0	0	0	0	0
13	12	H29425	H	298.6	309.1	1423	44.3	29	0	0	0	0	0	0
14	13	M14872	M	298.6	309.1	1339	51.1	34	0	0	0	0	0	0
15	14	M14873	M	298.6	309.2	1742	30	37	0	0	0	0	0	0
16	15	L47194	L	298.6	309.2	2035	19.6	40	0	0	0	0	0	0
17	16	L47195	L	298.6	309.2	1542	48.4	42	0	0	0	0	0	0
18	17	M14876	M	298.6	309.2	1311	46.6	44	0	0	0	0	0	0
19	18	M14877	M	298.7	309.2	1410	45.6	47	0	0	0	0	0	0
20	19	H29432	H	298.8	309.2	1306	54.5	50	0	0	0	0	0	0
21	20	M14879	M	298.9	309.3	1632	32.5	55	0	0	0	0	0	0
22	21	H29434	H	298.9	309.3	1375	42.7	58	0	0	0	0	0	0
23	22	L47201	L	298.8	309.3	1450	44.8	63	0	0	0	0	0	0
24	23	M14882	M	298.9	309.3	1581	30.7	65	0	0	0	0	0	0
25	24	L47203	L	299	309.4	1758	25.7	68	0	0	0	0	0	0
26	25	M14884	M	299	309.4	1561	37.3	70	0	0	0	0	0	0
27	26	L47205	L	299	309.5	1861	23.3	73	0	0	0	0	0	0
28	27	L47206	L	299.1	309.5	1512	39	75	0	0	0	0	0	0
29	28	H29441	H	299.1	309.4	1811	24.6	77	0	0	0	0	0	0
30	29	L47208	L	299.1	309.4		44.2		0	0	0	0	0	0
31	30	L47209		299	309.4	1693	30.1	84	0	0	0	0	0	0
32	31	M14890	M	299.1	309.5	1339	48.2	86	0	0	0	0	0	0
33	32	L47211		299	309.4	1798	25.5	89	0	0	0	0	0	0
34	33	L47212	L	299	309.4	1419	48.3	91	0	0	0	0	0	0
35	34	L47213		298.9	309.3	1665	32.5	93	0	0	0	0	0	0

2. CORE SENSOR PARAMETERS:

The dataset contains five numerical sensor parameters and one categorical product-type attribute. These parameters represent machine operating conditions and are used as inputs for fault prediction.

Core Sensor Parameters

Parameter	Unit	Description
Air Temperature	K	Ambient air temperature
Process Temperature	K	Temperature during machine operation
Rotational Speed	rpm	Rotational speed of the machine
Torque	Nm	Rotational force applied to the machine
Tool Wear	min	Accumulated tool wear time
Product Type	L/M/H	Product quality variant

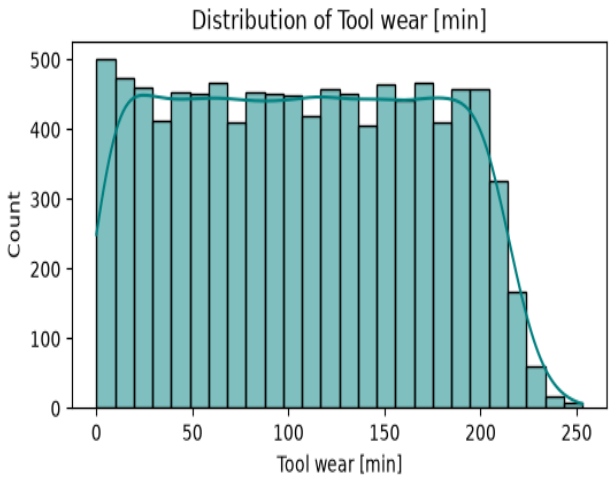
3. DATA EXPLORATION AND CLEANING:

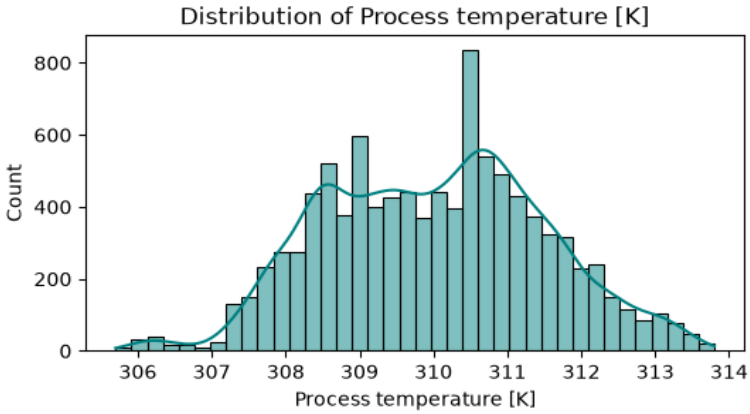
The dataset was loaded and inspected to examine its dimensions, data types, missing values and duplicate records. Exploratory Data Analysis (EDA) was performed using histograms, boxplots, correlation heatmaps and failure distribution charts to understand sensor patterns and identify potential outliers.

The messy dataset was further processed to handle duplicate records, missing machine types and invalid sensor values. The binary Machine Failure label was compared with the five failure-mode indicators to identify inconsistencies. The dataset was then divided into training and testing sets using an 80:20 stratified split. Missing numerical values were replaced using median values calculated from the training data.

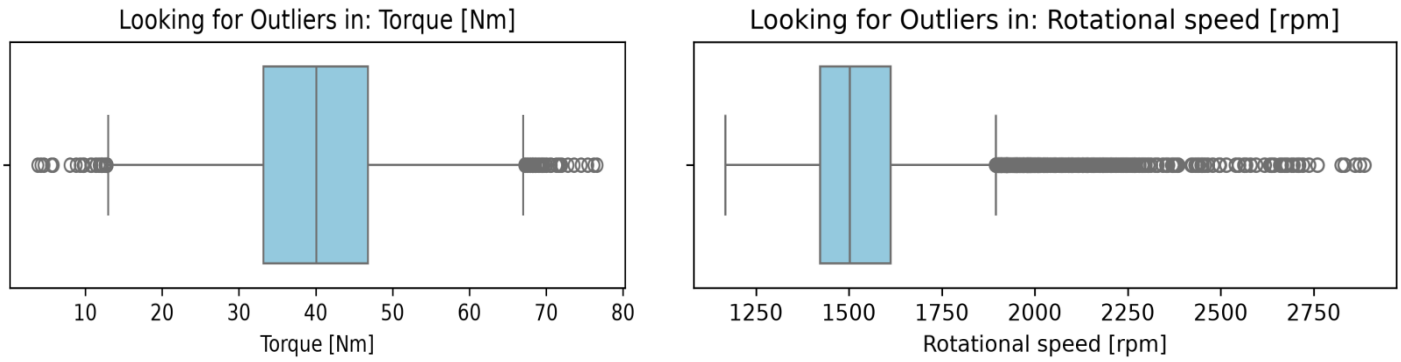
The following visualizations were used during data exploration:

- **Sensor Parameter Distributions (Histograms)**

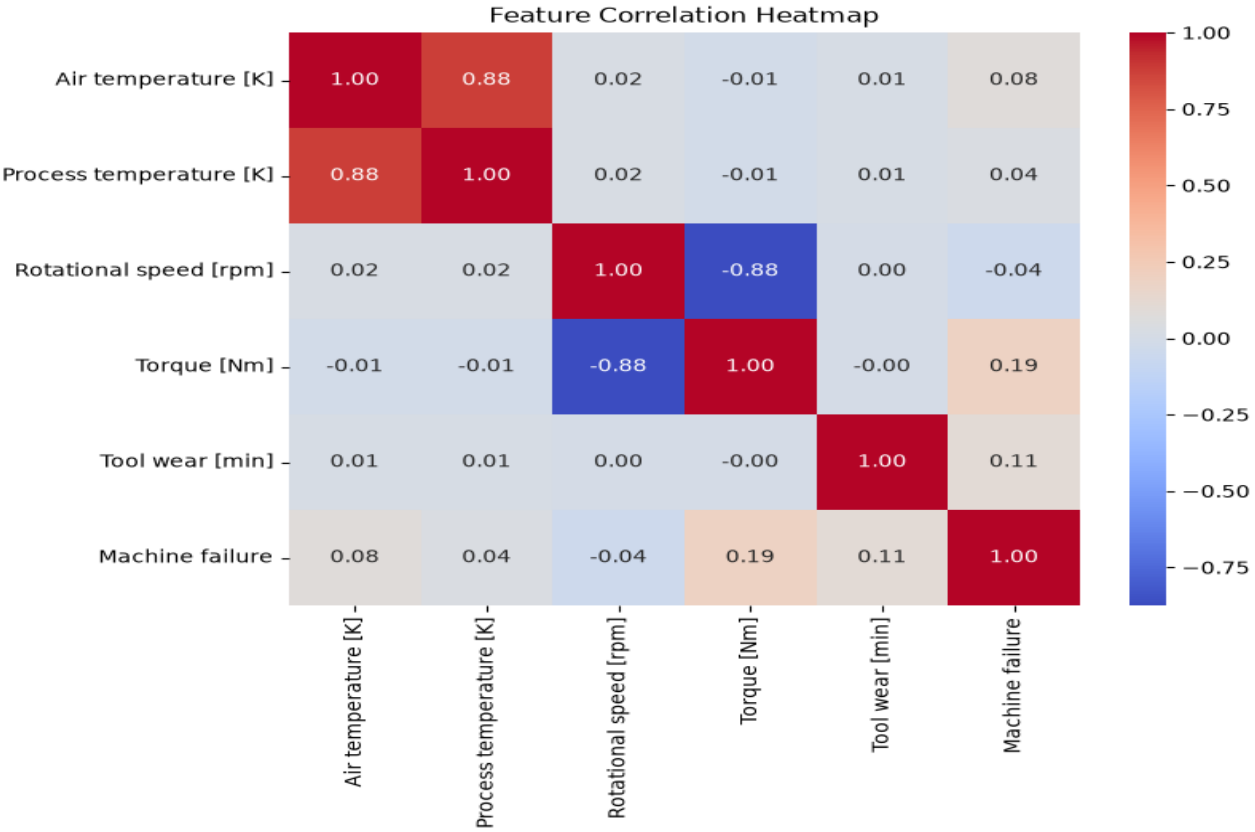




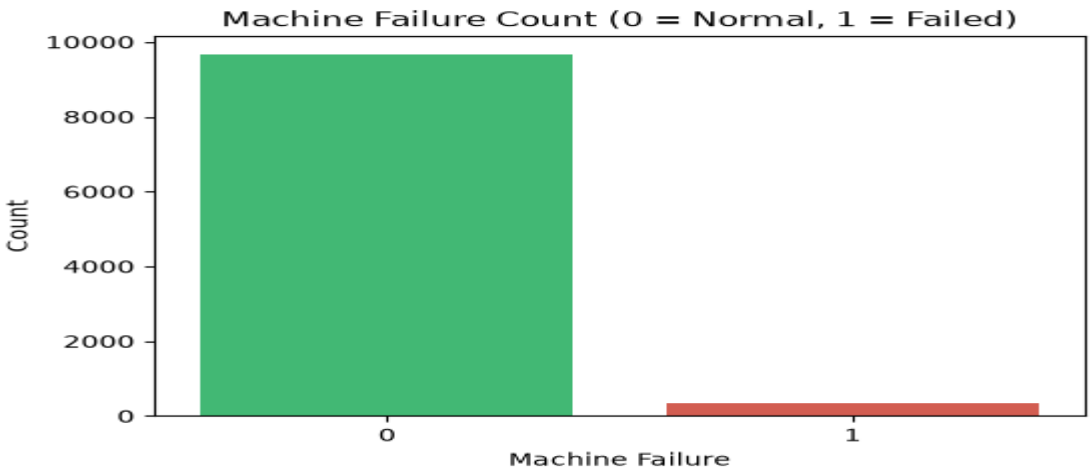
- **Outlier Analysis (Boxplots)**



- **Correlation Heatmap**



- **Machine Failure Class Distribution**



4. FEATURE ENGINEERING AND MODEL DEVELOPMENT:

Feature engineering was performed to derive additional information from the original sensor parameters and improve the model's ability to identify machine failure patterns.

Engineered Features:

Feature	Calculation
Temperature Difference	**Process Temperature – Air Temperature**
Mechanical Power	**Torque × Angular Velocity**
Overstrain Index	**Tool Wear × Torque**
Torque-Speed Ratio	**Torque ÷ Rotational Speed**

Machine Type was converted into numerical form using One-Hot Encoding. Feature scaling was applied to Logistic Regression, while tree-based models used unscaled engineered features.

Four machine-learning algorithms were evaluated: Logistic Regression, Random Forest, XGBoost and LightGBM. XGBoost was tested in both baseline and tuned configurations. Class imbalance was considered during model development. SMOTE-based oversampling and classification threshold adjustments were also experimented with; however, SMOTE did not improve the results.

XGBoost was further optimized using Randomized Search with five-fold stratified cross-validation and F1-Score as the optimization metric.

Model Performance Comparison:

	A	B	C	D	E	F	G
1	Model	Precision	Recall	F1-Score	ROC-AUC	False Positives	False Negatives
2	Logistic Regression	0.168115942	0.852941176	0.280871671	0.925960602	287	10
3	Random Forest	0.617283951	0.735294118	0.67114094	0.972133419	31	18
4	XGBoost	0.828125	0.779411765	0.803030303	0.948909999	11	15
5	LightGBM	0.8	0.764705882	0.781954887	0.947128852	13	16

The tuned XGBoost model achieved the highest F1-Score and was selected as the final model. It was saved and integrated into the FastAPI backend for machine-failure prediction.

5. KNOWLEDGE SOURCES:

Along with the predictive dataset, FaultSense AI uses a knowledge base to support rule-based diagnosis and generate maintenance recommendations. It contains fault definitions, affected components, machine-type-specific limits, technical references and maintenance protocols for different stakeholders. The knowledge representation and associated diagnostic rules provide additional domain knowledge to complement the machine-learning predictions.

Knowledge Base Information:

Knowledge Component	Purpose
Fault Definitions	Describes the five failure categories: TWF, HDF, PWF, OSF, and RNF.
Knowledge Representation	Organizes fault-related knowledge and relationships for use in diagnosis.
Affected Components	Identifies machine components associated with different faults.
Machine-Type Limits	Stores diagnostic limits for L, M, and H product types.
ISO References	Provides supporting technical and safety references.
Maintenance Protocols	Provides recommendations for operators, technicians, and managers.

Implementation:

1. SOFTWARE / TOOLS / LIBRARIES:

FaultSense AI is developed using a combination of frontend, backend, machine learning, and deployment technologies. The following tools and libraries are used to build the system:

Category	Tools / Libraries	Purpose
Backend Framework	**FastAPI, Uvicorn, Pydantic**	**FastAPI handles API requests and diagnosis processing, Uvicorn runs the application, and Pydantic validates input data.**
Machine Learning & Analytics	**Python 3.10+, Scikit-Learn, XGBoost, LightGBM, Pandas, NumPy, Joblib**	**Used for data preprocessing, feature engineering, model training, evaluation, prediction, and loading trained model artifacts.**
Frontend Dashboard	**React.js, JSX, CSS3, Lucide-React, Vite**	**Used to develop the interactive user interface, input forms, diagnostic results, and responsive dashboard.**
Deployment Platform	**Vercel**	**Used to deploy the serverless Python backend and static React frontend.**

2. MODULES AND COMPONENTS:

FaultSense AI is divided into several modules, each responsible for a specific part of the diagnosis process.

- **InputPage.jsx – Frontend Telemetry Interface**

Developed using React.js and JSX, this component provides the main user input interface. It includes numerical input fields, sliders, and product type selectors (L, M, H) to collect machine sensor readings. The entered values are organized into a structured JSON payload and sent to the backend API for diagnosis.

- **main.py – FastAPI Application Core**

This file serves as the main backend application. It provides the REST API endpoint for receiving sensor data, handles incoming HTTP requests, configures CORS middleware, and connects the frontend with the diagnostic system. It also supports the backend deployment configuration.

- **knowledge_base.py – Symbolic Knowledge Repository**

This module contains the centralized KNOWLEDGE_BASE dictionary. It stores information about industrial failure modes (TWF, HDF, PWF, OSF, and RNF), affected machine components, product-type-specific limits, ISO standard references, and maintenance action protocols.

- **inference_engine.py – Hybrid Inference Core**

This module manages input validation, calculates derived machine parameters, and performs fault diagnosis. It combines machine learning predictions from the trained XGBoost model with symbolic forward-chaining rules to identify potential machine faults.

- **explaination_generator.py – Diagnostic Explanation Generator**

This module converts the inference results and sensor information into structured, human-readable diagnostic reports. The explanations cover five analytical pillars: What, Why, How Sure, Impact, and Action Protocols.

- **notebooks/ and models/ – Machine Learning Pipeline and Artifacts**

The notebooks directory contains the Jupyter Notebook used for exploratory data analysis, data preprocessing, feature engineering, and model development. The models directory stores the trained model and supporting artifacts, including best_xgb_tuned.joblib, scaler.joblib, and feature_names.joblib, which are used during prediction.

3. CODE STRUCTURE:

The project is organized into separate directories for the backend, frontend, machine learning models, datasets, and documentation. This modular structure makes the system easier to maintain and develop.

```
kbs-industrial-fault-diagnosis/
│
├── main.py
├── requirements.txt
├── vercel.json
├── README.md
│
├── models/
│   ├── best_xgb_tuned.joblib
│   ├── scaler.joblib
│   └── feature_names.joblib
│
├── backend/
│   ├── inference_engine.py
│   ├── knowledge_base.py
│   ├── explaination_generator.py
│   └── kbs_project_data_pipline..ipynb
│
├── data/
│   ├── raw/
│   │   └── ai4i2020_messy.csv
│   └── processed/
│
├── docs/
```

```
└── frontend/
    ├── package.json
    ├── vite.config.js
    ├── vercel.json
    ├── .env.example
    │
    └── src/
        ├── App.jsx
        ├── services/
        │   └── diagnosisApi.js
        └── components/
            ├── Hero.jsx
            ├── InputPage.jsx
            └── DiagnosticResult.jsx
```

4. USER INPUTS AND OUTPUTS:

FaultSense AI allows users to enter machine operating parameters through an interactive dashboard. After submission, the system processes the readings and provides a fault diagnosis with supporting explanations and recommended actions.

User Inputs

The system accepts the following machine parameters:

Input Parameter	Unit / Options	Description
Air Temperature	Kelvin (K)	Temperature of the surrounding air.
Process Temperature	Kelvin (K)	Temperature of the machine process.
Rotational Speed	RPM	Rotational speed of the machine.
Torque	Nm	Torque applied by the machine.
Tool Wear	Minutes	Accumulated tool wear time.
Product Type	L, M, H	Machine product variant.

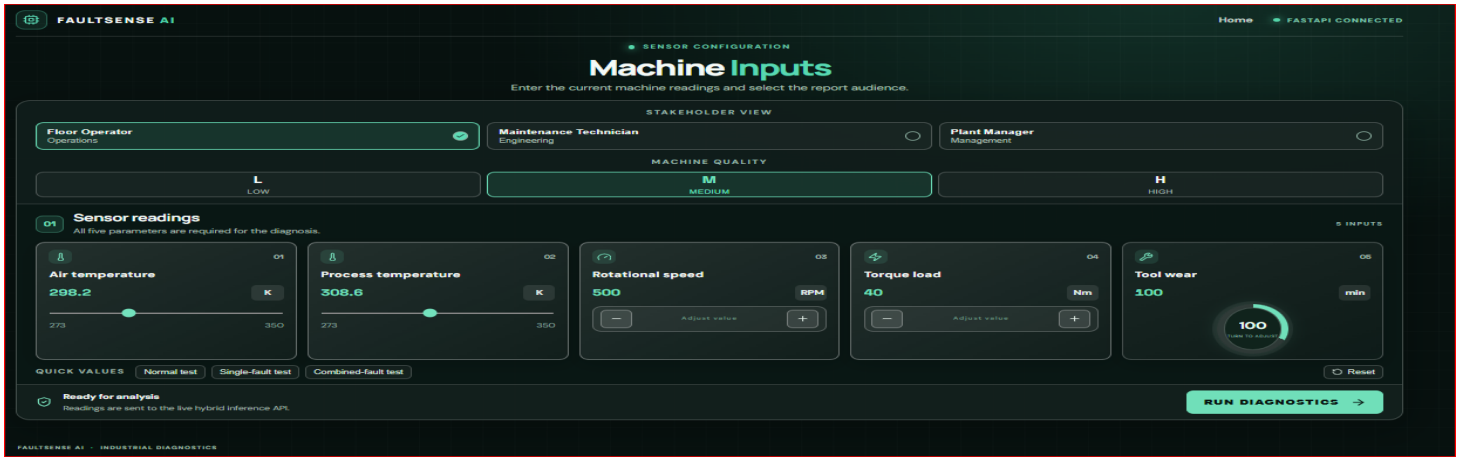
Users manually enter the required sensor readings and select the product type through the dashboard. The submitted values are sent to the backend for processing and diagnosis.

System Outputs

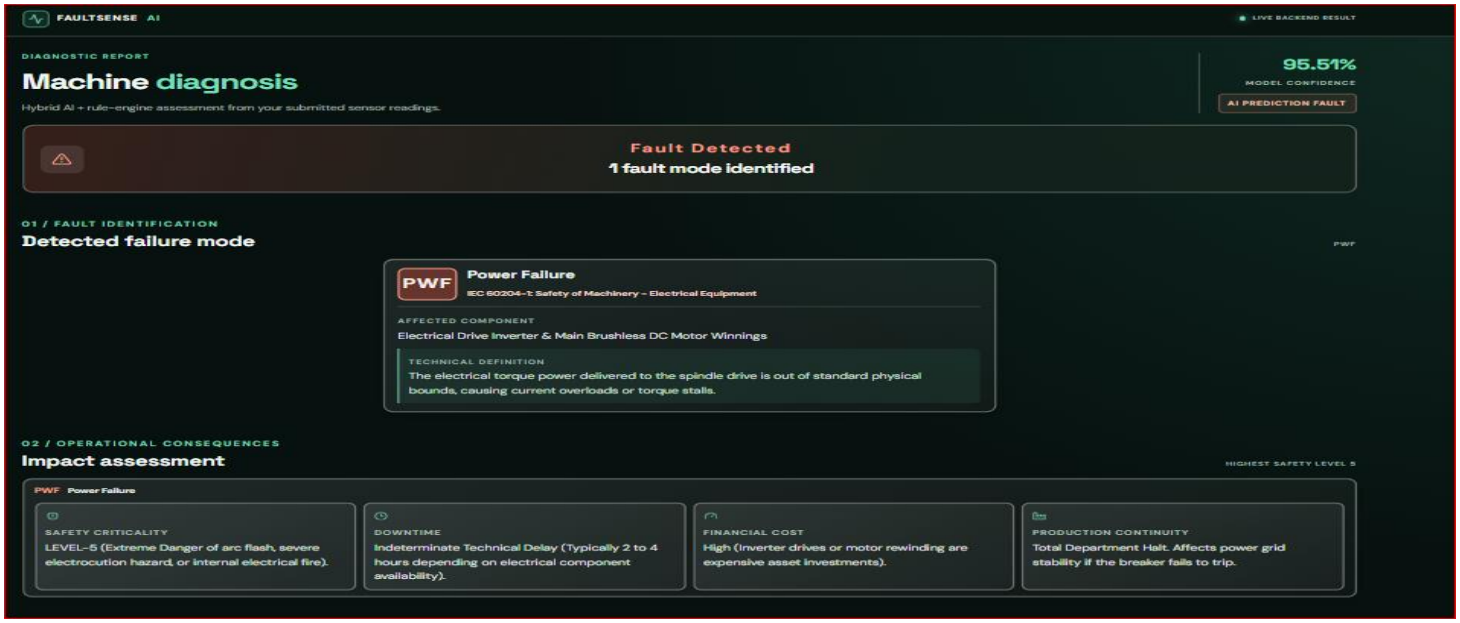
After processing the submitted inputs, FaultSense AI provides:

- **Machine Status:** Indicates whether the machine is operating normally or a fault has been detected.
- **Predicted Fault Category:** Identifies the predicted failure mode, such as TWF, HDF, PWF, OSF, or RNF.
- **ML Confidence Score:** Displays the confidence score returned by the machine learning model.
- **Rule-Based Detection:** Shows the faults identified by the symbolic rules.
- **Diagnostic Explanation:** Provides details about what the fault is, why it may have occurred, and how confident the model is.
- **Impact Assessment:** Describes the possible effect of the detected fault on machine operation.
- **Recommended Actions:** Provides maintenance guidance and relevant action protocols.

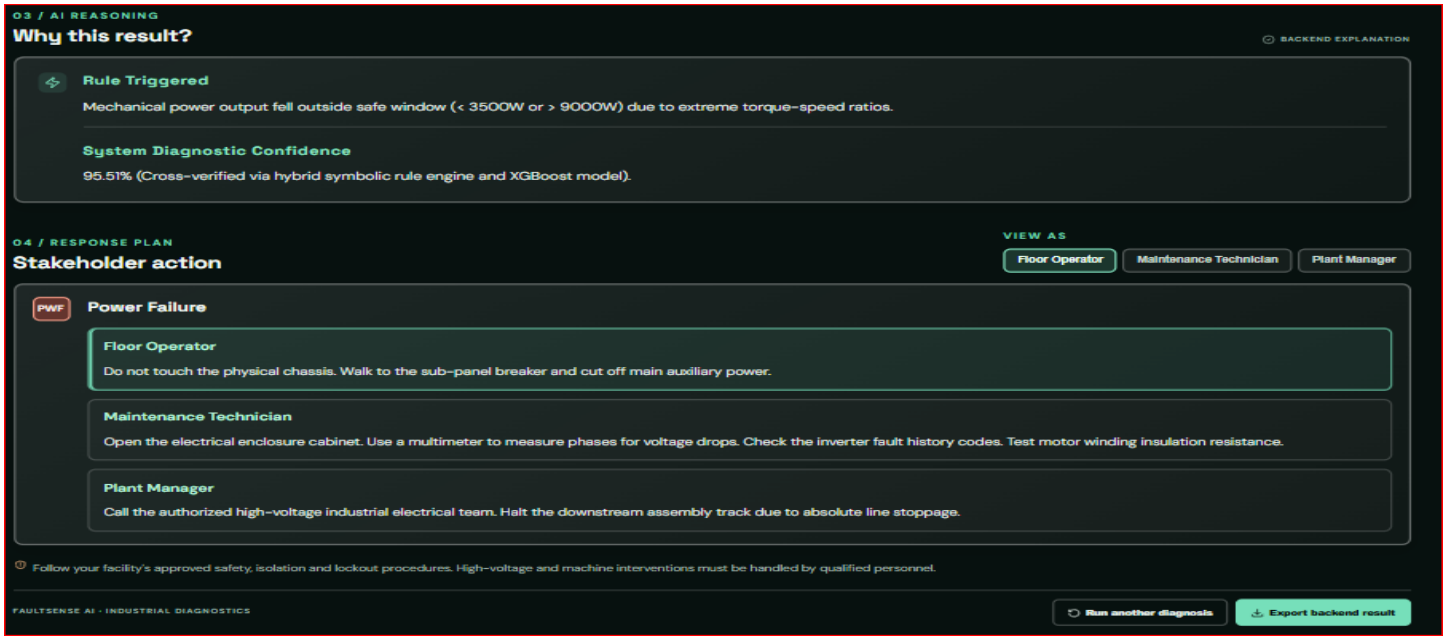
User Input Interface



Diagnostic Results:



Fault Explanation and Recommended Actions:



TESTING AND RESULTS:

1. TESTING APPROACH:

FaultSense AI was tested using different machine sensor readings to evaluate normal operation, single-fault detection, and multiple-fault detection. Input validation and diagnostic outputs were also considered during testing.

2. TEST CASES:

The following test cases were performed using Product Type **M**.

Test Case	Air Temp (K)	Process Temp (K)	RPM	Torque (Nm)	Tool Wear (min)	Expected Condition
TC-01	300	310	1800	20	50	Normal
TC-02	300	305	1800	20	230	Single Fault – TWF
TC-03	300	301	1200	80	180	Multiple Faults – HDF, PWF and OSF

The system was evaluated using the three test cases. The following screenshots show the diagnostic outputs generated for each condition.

Normal Machine Diagnosis (TC-01)

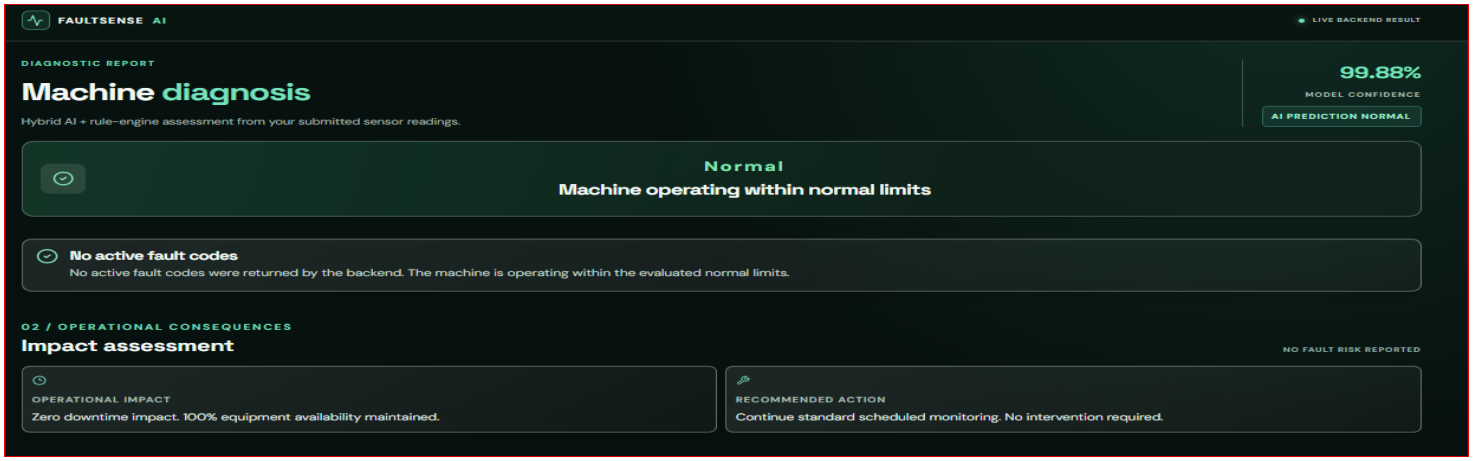
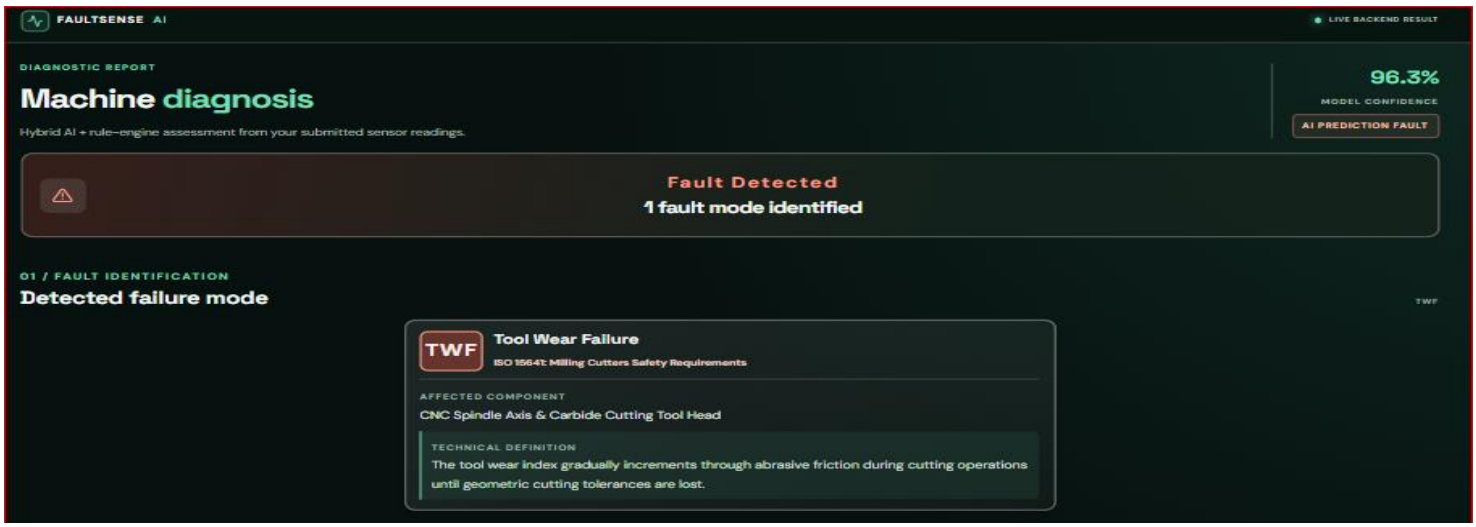
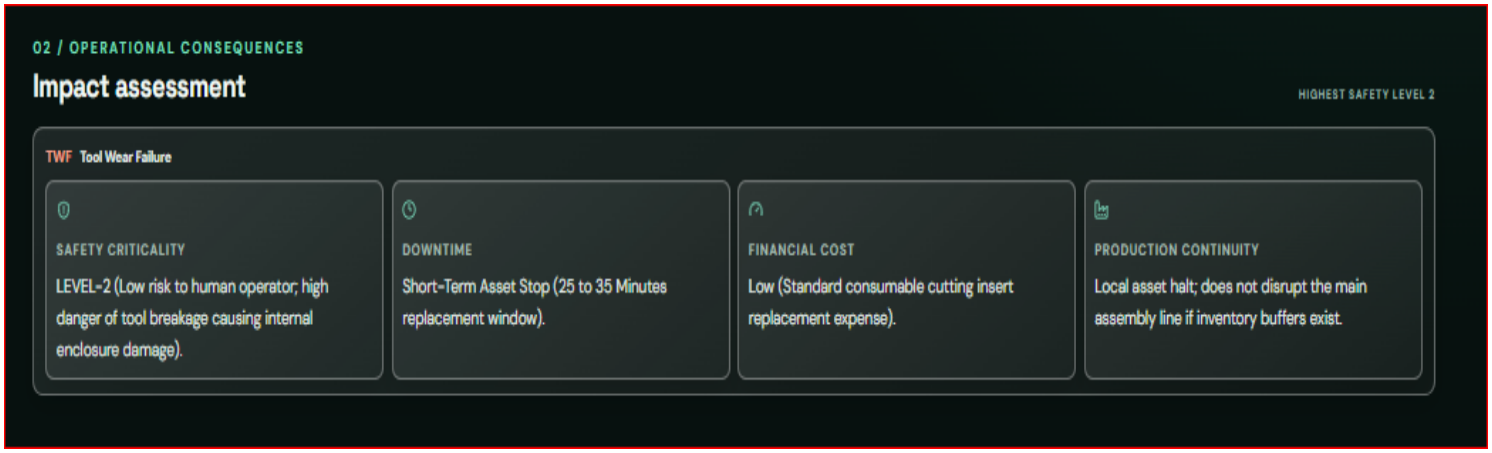
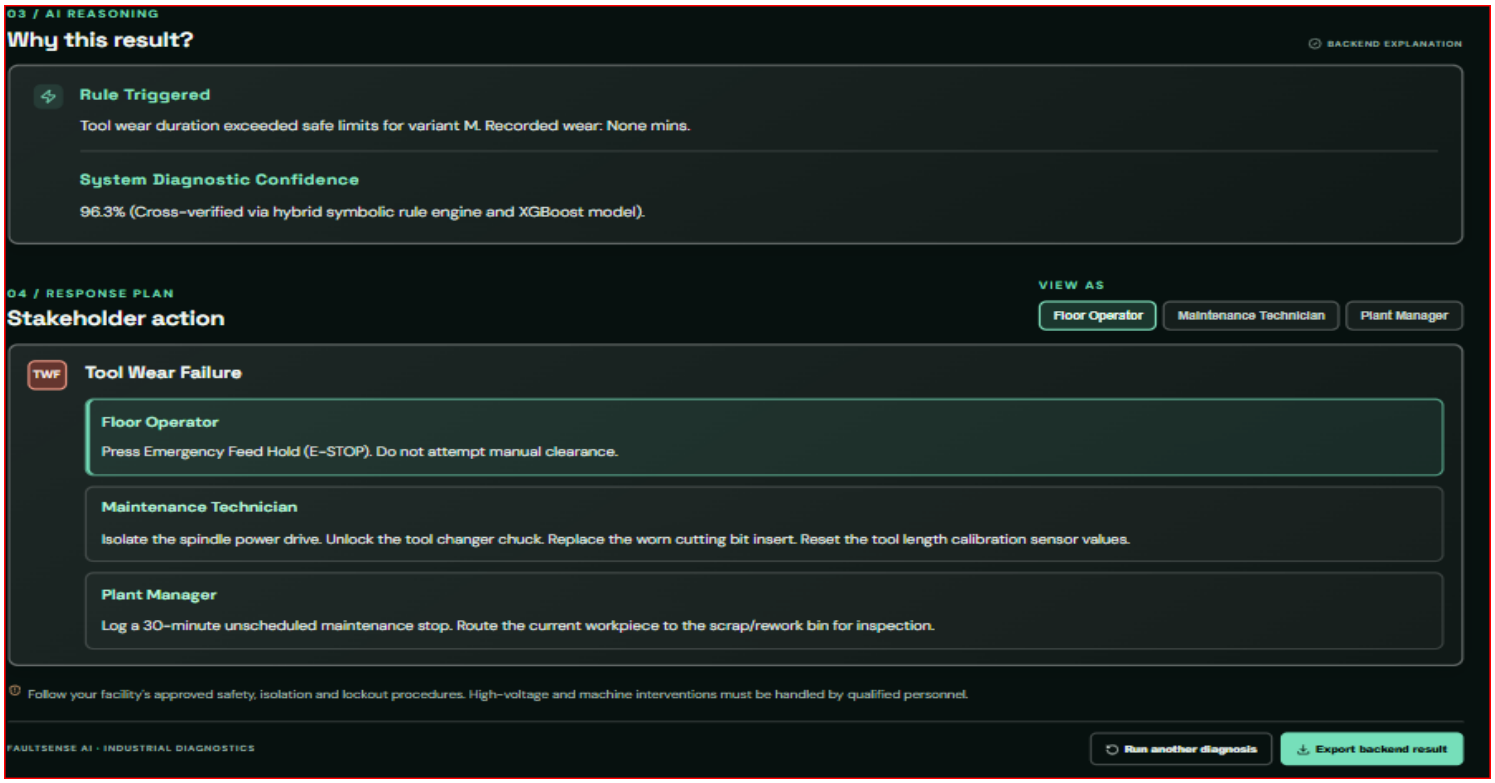


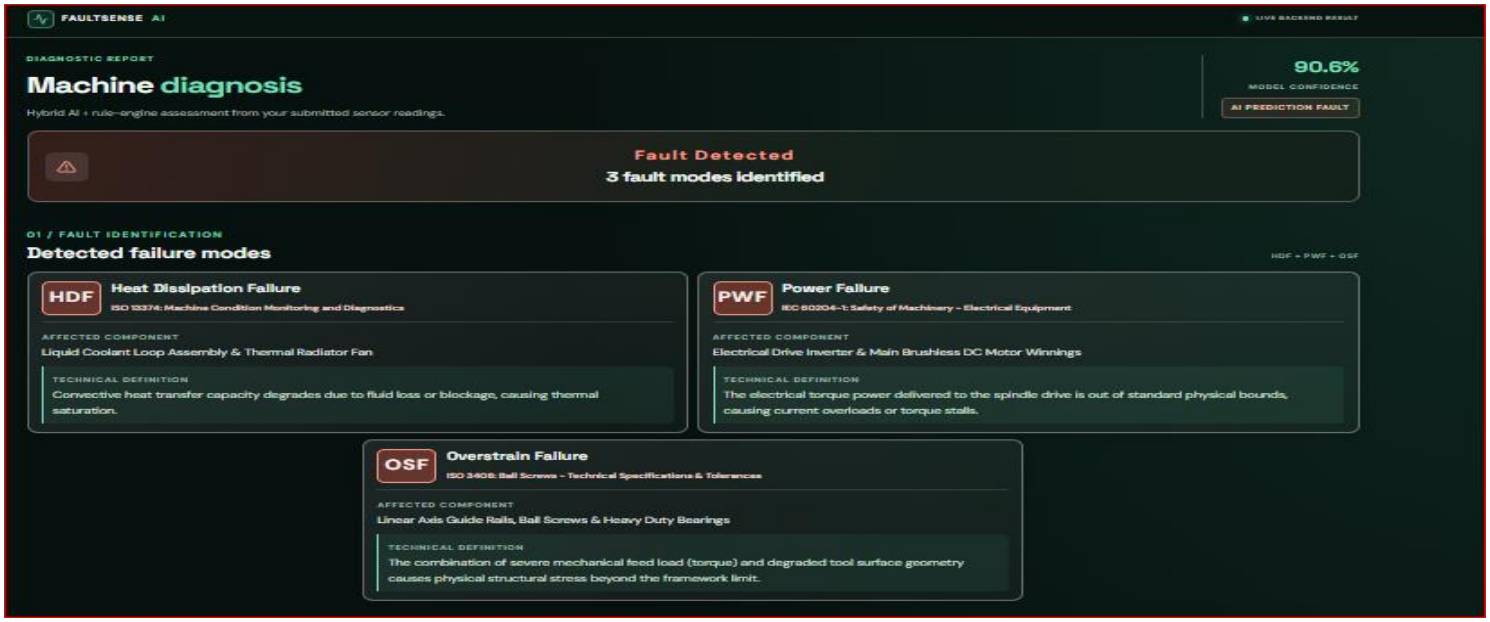
Figure 7.2: Single Fault Detection – TWF (TC-02)

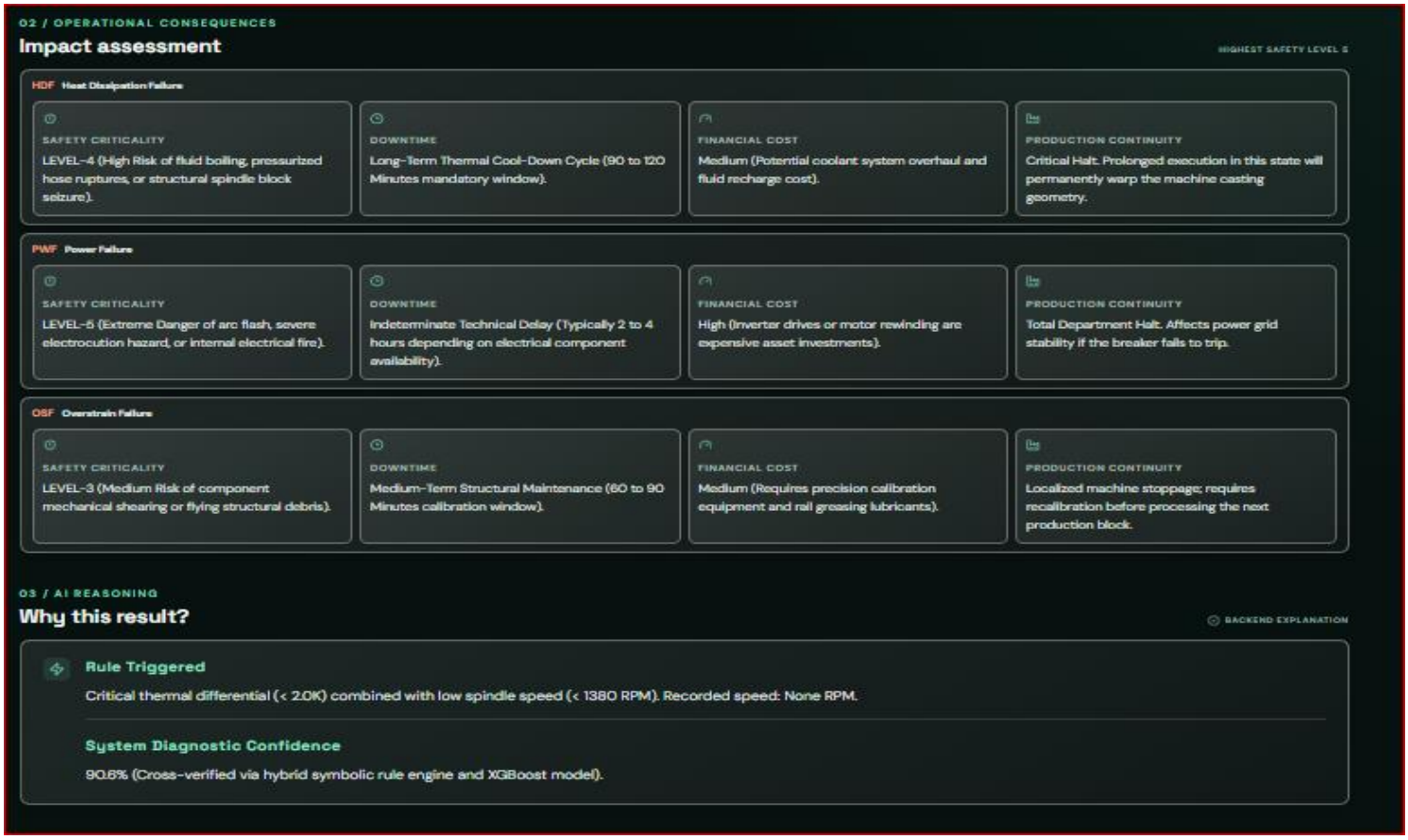


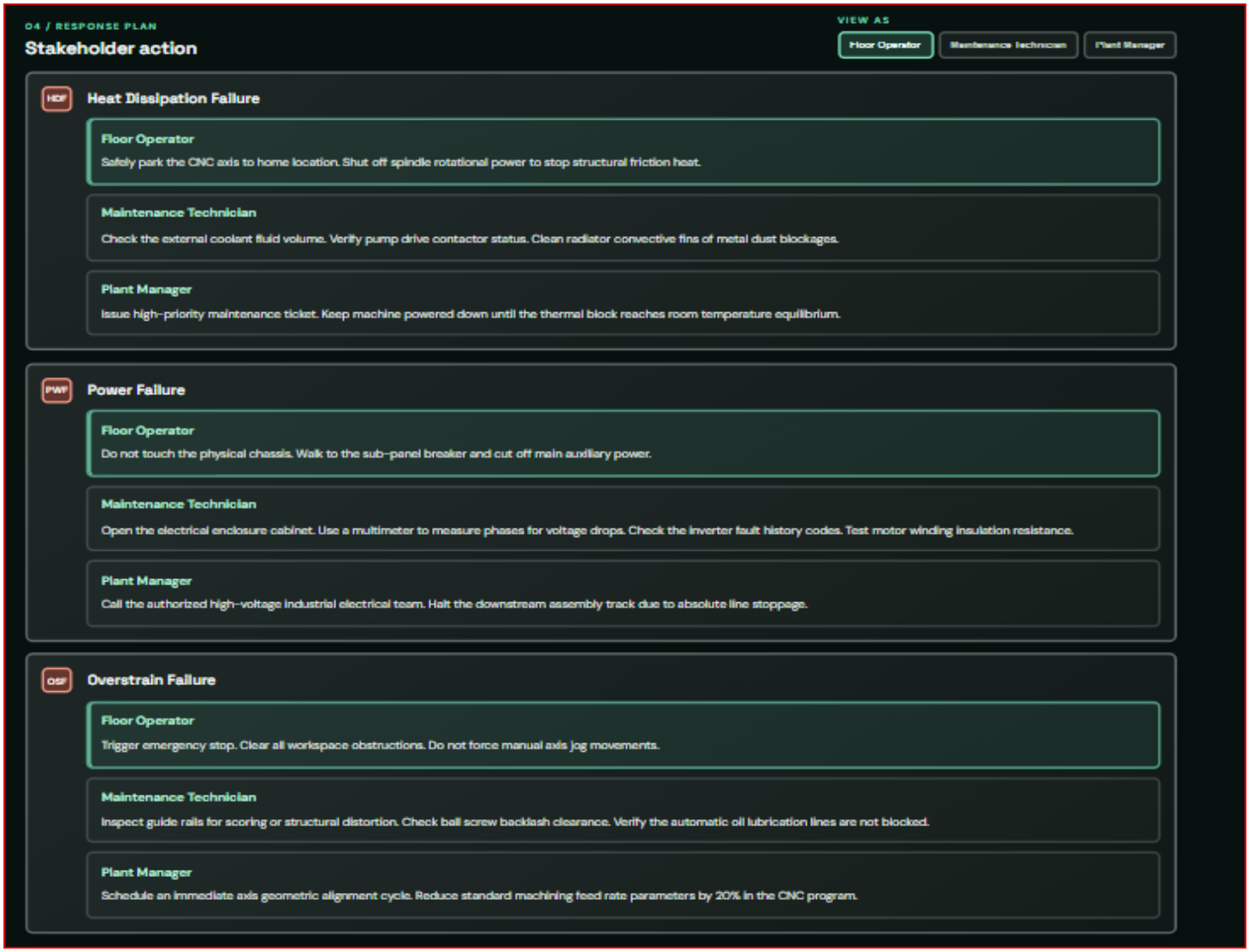




Multiple Fault Detection (TC-03)




02 / OPERATIONAL CONSEQUENCES
Impact assessment
HIGHEST SAFETY LEVEL 5
HDF Heat Dissipation Failure
SAFETY CRITICALITY
LEVEL-4 (High Risk of fluid boiling, pressurized hose ruptures, or structural spindle block seizure).
DOWNTIME
Long-Term Thermal Cool-Down Cycle (90 to 120 Minutes mandatory window).
FINANCIAL COST
Medium (Potential coolant system overhaul and fluid recharge cost).
PRODUCTION CONTINUITY
Critical Halt. Prolonged execution in this state will permanently warp the machine casting geometry.
PWF Power Failure
SAFETY CRITICALITY
LEVEL-5 (Extreme Danger of arc flash, severe electrocution hazard, or internal electrical fire).
DOWNTIME
Indeterminate Technical Delay (Typically 2 to 4 hours depending on electrical component availability).
FINANCIAL COST
High (Inverter drives or motor rewinding are expensive asset investments).
PRODUCTION CONTINUITY
Total Department Halt. Affects power grid stability if the breaker fails to trip.
OSF Overstrain Failure
SAFETY CRITICALITY
LEVEL-3 (Medium Risk of component mechanical shearing or flying structural debris).
DOWNTIME
Medium-Term Structural Maintenance (60 to 90 Minutes calibration window).
FINANCIAL COST
Medium (Requires precision calibration equipment and rail greasing lubricants).
PRODUCTION CONTINUITY
Localized machine stoppage; requires recalibration before processing the next production block.
03 / AI REASONING
Why this result?
BACKEND EXPLANATION
Rule Triggered
Critical thermal differential (< 2.0K) combined with low spindle speed (< 1380 RPM). Recorded speed: None RPM.
System Diagnostic Confidence
90.6% (Cross-verified via hybrid symbolic rule engine and XGBoost model).


04 / RESPONSE PLAN
Stakeholder action
VIEW AS
Floor Operator
Maintenance Technician
Plant Manager
HDF
Heat Dissipation Failure
Floor Operator
Safely park the CNC axis to home location. Shut off spindle rotational power to stop structural friction heat.
Maintenance Technician
Check the external coolant fluid volume. Verify pump drive contactor status. Clean radiator convective fins of metal dust blockages.
Plant Manager
Issue high-priority maintenance ticket. Keep machine powered down until the thermal block reaches room temperature equilibrium.
PWF
Power Failure
Floor Operator
Do not touch the physical chassis. Walk to the sub-panel breaker and cut off main auxiliary power.
Maintenance Technician
Open the electrical enclosure cabinet. Use a multimeter to measure phases for voltage drops. Check the inverter fault history codes. Test motor winding insulation resistance.
Plant Manager
Call the authorized high-voltage industrial electrical team. Halt the downstream assembly track due to absolute line stoppage.
OSF
Overstrain Failure
Floor Operator
Trigger emergency stop. Clear all workspace obstructions. Do not force manual axis jog movements.
Maintenance Technician
Inspect guide rails for scoring or structural distortion. Check ball screw backlash clearance. Verify the automatic oil lubrication lines are not blocked.
Plant Manager
Schedule an immediate axis geometric alignment cycle. Reduce standard machining feed rate parameters by 20% in the CNC program.

3. JSON OUTPUT:


```
{
  "ml_prediction_label": 1,
  "status": "Fault Detected",
  "confidence_probability": 96.3,
  "detected_fault_codes": [
    "TWF"
  ],
  "explanation": "Verified active anomalies: TWF",
  "fault_details_map": {
    "TWF": {
      "failure_mode": "Tool Wear Failure",
      "target_component": "CNC Spindle Axis & Carbide Cutting Tool Head",
      "iso_standard": "ISO 15641: Milling Cutters Safety Requirements",
      "technical_definition": "The tool wear index gradually increments through abrasive friction during cutting operations until geometric cutting tolerances are lost.",
      "action_protocols": {
        "Floor_Operator_Action": "Press Emergency Feed Hold (E-STOP). Do not attempt manual clearance.",
        "Maintenance_Technician_Action": "Isolate the spindle power drive. Unlock the tool changer chuck. Replace the worn cutting bit insert. Reset the tool length calibration sensor values.",
        "Plant_Manager_Action": "Log a 30-minute unscheduled maintenance stop. Route the current workpiece to the scrap/rework bin for inspection."
      },
      "constraints_matrix": {
        "Safety_Criticality": "LEVEL-2 (Low risk to human operator; high danger of tool breakage causing internal enclosure damage).",
        "Downtime_Impact": "Short-Term Asset Stop (25 to 35 Minutes replacement window).",
        "Financial_Cost_Index": "Low (Standard consumable cutting insert replacement expense).",
        "Production_Continuity": "Local asset halt; does not disrupt the main assembly line if inventory buffers exist."
      }
    }
  },
  "explanation_report": {
    "summary_card": "DIAGNOSTIC REPORT: Active Fault Mode(s) Detected: TWF (Tool Wear Failure) affecting the CNC Spindle Axis & Carbide Cutting Tool Head. Rule Triggered: Tool wear duration exceeded safe limits for variant M. Recorded wear: 230 mins. System Diagnostic Confidence: 96.3% (Cross-verified via hybrid symbolic rule engine and XGBoost model). Operational Impact — Downtime: Short-Term Asset Stop (25 to 35 Minutes replacement window). | Financial Risk: Low (Standard consumable cutting insert replacement expense). | Safety Level: LEVEL-2 (Low risk to human operator; high danger of tool breakage causing internal enclosure damage). Immediate Protocol: Press Emergency Feed Hold (E-STOP). Do not attempt manual clearance.",
    "what": "Active Fault Mode(s) Detected: TWF (Tool Wear Failure) affecting the CNC Spindle Axis & Carbide Cutting Tool Head.",
    "why": "Rule Triggered: Tool wear duration exceeded safe limits for variant M. Recorded wear: 230 mins.",
    "how_sure": "System Diagnostic Confidence: 96.3% (Cross-verified via hybrid symbolic rule engine and XGBoost model).",
    "impact": "Operational Impact — Downtime: Short-Term Asset Stop (25 to 35 Minutes replacement window). | Financial Risk: Low (Standard consumable cutting insert replacement expense). | Safety Level: LEVEL-2 (Low risk to human operator; high danger of tool breakage causing internal enclosure damage).",
```

```
  "action": "Recommended Action Protocols -> [Operator]: Press Emergency Feed Hold (E-STOP). Do not attempt manual clearance. | [Technician]: Isolate the spindle power drive. Unlock the tool changer chuck. Replace the worn cutting bit insert. Reset the tool length calibration sensor values. | [Manager]: Log a 30-minute unscheduled maintenance stop. Route the current workpiece to the scrap/rework bin for inspection."
 },
 "sensor_readings": {
  "air_temperature": 300,
  "process_temperature": 305,
  "rotational_speed": 1800,
  "torque": 20,
  "tool_wear": 230,
  "machine_type": "M"
 },
 "selected_role": "Floor Operator",
 "received_at": "2026-11-02T23:06:20.812Z" }
```


4. DIAGNOSTIC OUTPUT EXPLANATION:

FaultSense AI generates a structured diagnostic report to help users understand the detected machine condition, its possible causes, operational impact, and recommended actions.

- **Status:** Shows whether a machine is operating normally or a fault has been detected.
- **Confidence Probability:** Displays the machine learning model's confidence in its diagnostic prediction as a percentage.
- **Detected Fault Codes:** Lists the fault codes identified by the system, such as HDF, PWF, and OSF. Multiple faults can be detected together.
- **Explanation:** Provides a brief summary of the anomalies identified by the hybrid system.
- **Failure Mode:** Identifies the type of failure detected, such as Heat Dissipation Failure, Power Failure, or Overstrain Failure.
- **Target Component:** Identifies the specific machine component associated with the detected fault.
- **ISO Standard:** Specifies the industrial standard associated with the detected fault.
- **Technical Definition:** Explains the technical cause and nature of the identified failure.
- **Action Protocols:** Provides recommended actions for three responsible roles:
  1. **Floor Operator Action:** Immediate steps to ensure machine and workplace safety.
  2. **Maintenance Technician Action:** Technical inspection and troubleshooting procedures.
  3. **Plant Manager Action:** Maintenance decisions and production management actions.
- **Constraints Matrix:** Describes the operational consequences and limitations associated with the fault:
  1. **Safety Criticality:** Indicates the level of risk to personnel and equipment.
  2. **Downtime Impact:** Estimates the time required for maintenance or recovery.
  3. **Financial Cost Index:** Indicates the expected maintenance or repair cost.
  4. **Production Continuity:** Explains how the fault affects ongoing production.
- **Summary Card:** Combines the detected fault, affected component, triggering condition, confidence, operational impact, and immediate action into one diagnostic summary.
- **What:** Describes which fault was detected and which component is affected.
- **Why:** Explains the sensor conditions or rule-based thresholds that triggered the diagnosis.
- **How Sure:** Shows the system's confidence in the diagnostic result and mentions its verification through the ML model and rule engine.
- **Impact:** Summarizes the fault's safety risk, expected downtime, and financial consequences.
- **Action:** Combines the recommended actions for the operator, technician, and manager.

EVALUATION, LIMITATIONS AND CONCLUSION:

1. EVALUATION OF THE PROPOSED APPROACH:

FaultSense AI was evaluated based on its machine learning performance, explainability, safety awareness, computational requirements, completeness and usability.

Criterion	Assessment
Model Performance	The tuned XGBoost model achieved a precision of 0.857, recall of 0.794, F1-score of 0.824 and ROC-AUC of 0.965. Rule-based reasoning adds specific fault detection based on predefined sensor conditions.
Explainability	Each detected fault is associated with a defined rule, relevant sensor conditions, technical references and a structured explanation covering What, Why, How Sure, Impact and Action.
Safety	The system highlights safety-related information and provides recommended actions for operators, maintenance technicians and plant managers. It also includes warnings about following appropriate safety and lockout procedures.
Computational Cost	The system uses a lightweight, CPU-based implementation with open-source tools. It does not require specialised GPU hardware for normal inference.
Completeness	The approach combines machine learning, a structured knowledge base, forward-chaining rules, diagnostic reasoning, conflict handling and explanation generation.
Usability	A web-based interface allows users to enter machine sensor readings and view diagnostic results. The backend also provides an API for communication with the frontend.

2. LIMITATIONS:

Overall, the proposed hybrid approach combines the predictive capability of machine learning with the transparent reasoning of predefined rules, making the diagnostic results more informative for maintenance-related decisions.

Despite its functionality, FaultSense AI has the following limitations:

- The diagnostic thresholds are predefined and may require adjustment for different machines and operating conditions.
- The confidence score represents the XGBoost model's predicted class probability. It is not a calibrated confidence score for the final rule-derived fault identity.
- The RNF category acts as a fallback when the ML model predicts a failure but no specific fault rule matches. Therefore, its explanation is more general than those of other fault categories.
- The knowledge base is manually defined and currently covers five failure categories. It does not automatically learn from maintenance history.
- Sensor readings are entered manually through the interface. The system does not currently receive live data directly from PLC, SCADA, or IoT devices.
- The system does not currently include persistent maintenance history or user authentication.
- Safety, downtime, and cost-related information is based on predefined estimates rather than actual site-specific industrial measurements.
- The rule thresholds and corresponding knowledge-base values must be maintained consistently to prevent differences between the values used for rule execution and those stored in the knowledge base.
- The system has been developed and evaluated using the AI4I 2020 synthetic dataset. Its performance has not yet been validated using real-world industrial machine data.
- The dataset contains a class imbalance, with normal records making up 96.61% and failure records 3.39%. This may affect the model's performance when applied to unseen real-world data.

3. FUTURE IMPROVEMENTS:

The following improvements can enhance the reliability and practical use of FaultSense AI:

- Validate the system using real-world CNC machine data and feedback from industrial maintenance experts.
- Store diagnostic thresholds in a single knowledge-base source and introduce machine-specific threshold configuration.
- Expand the knowledge base with additional machine types, components, symptoms, causes and maintenance actions.
- Develop a more advanced knowledge representation, such as an ontology or knowledge graph, to represent relationships between equipment, components, symptoms, causes and recommended actions.
- Improve input validation by clearly reporting missing, invalid or substituted sensor readings.
- Add degraded-mode warnings and more reliable handling of machine learning model errors.
- Calibrate model probabilities and introduce SHAP-based explanations to provide additional insight into machine learning predictions.
- Integrate live sensor readings through PLC, SCADA or IoT systems for continuous machine monitoring.
- Introduce trend-based early fault warnings, remaining useful life prediction and maintenance-history tracking.
- Add user authentication, audit logs and a maintenance-priority mechanism based on safety, cost, downtime and production continuity.

4. CONCLUSION:

FaultSense AI demonstrates a hybrid approach to industrial machine fault diagnosis by combining a tuned XGBoost model with a structured knowledge base and forward-chaining rules. The system analyses machine sensor readings, identifies potential failure conditions and generates explanations with relevant technical references and recommended maintenance actions.

The tuned XGBoost model achieved an F1-score of 0.824 and a ROC-AUC of 0.965 on the evaluated dataset. The rule-based component complements the machine learning prediction by identifying specific fault conditions through predefined sensor thresholds. The system also presents diagnostic information in a structured format for operators, maintenance technicians and plant managers.

Although the current implementation is limited by synthetic data, predefined thresholds and manual sensor input, it provides a foundation for an explainable industrial fault diagnosis system. Overall, FaultSense AI demonstrates the application of machine learning, knowledge representation and rule-based reasoning to a practical industrial maintenance problem.

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